

## Lab 1 — Setting up Google Colab

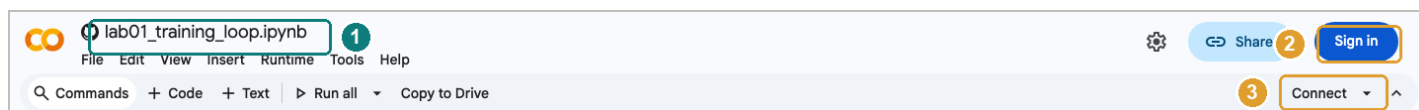
Everything runs in your browser. Nothing to install.

**What you need:** a web browser and a **free Google account** (a personal @gmail.com is fine — some hospital and university accounts block Colab, so if yours does, use a personal one). **You do not need:** Python, PyTorch, an install, or a GPU. *Colab* is a free Google service that runs Python notebooks on Google's computers; your laptop only shows the page.

### Step 1 — Open the Lab 1 notebook

Click the Lab 1 link (on the course page, or type it in). It opens Colab with the notebook already loaded — there is nothing to download first.

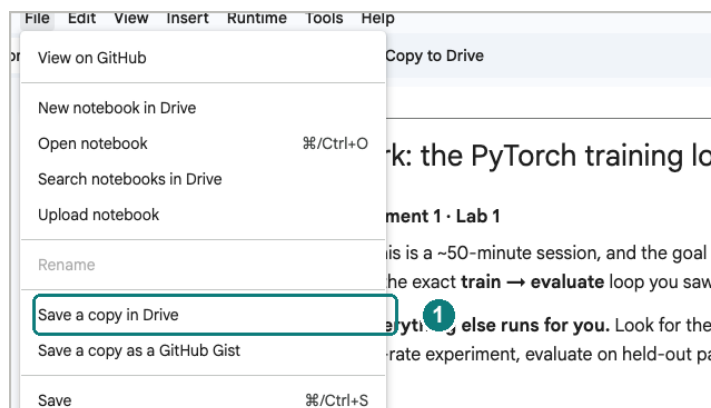
[https://colab.research.google.com/github/indigobio/msacl\\_dl\\_course/blob/main/student\\_pack/labs/lab01\\_training\\_loop.ipynb](https://colab.research.google.com/github/indigobio/msacl_dl_course/blob/main/student_pack/labs/lab01_training_loop.ipynb)



- (1) Check the file name reads `lab01_training_loop.ipynb` — that is the right notebook.
- (2) **Sign in** (top right) with your Google account. Until you do, you can read the notebook but not run it.
- (3) **Connect** (top right) starts your free machine. It usually connects on its own the first time you run a cell; if it says *Connect*, click it and wait for a green tick.

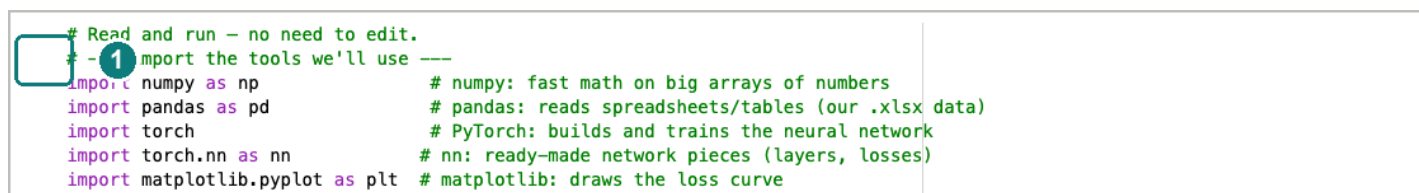
### Step 2 — Save your own copy (do this before you type anything)

The link opens a *read-only* copy. Use **File** → **Save a copy in Drive** and work in that copy, or your edits will be lost when you close the tab. Colab opens the copy in a new tab, named *Copy of lab01\_training\_loop.ipynb* — **that is the tab you want to work in**.



### Step 3 — Run a cell

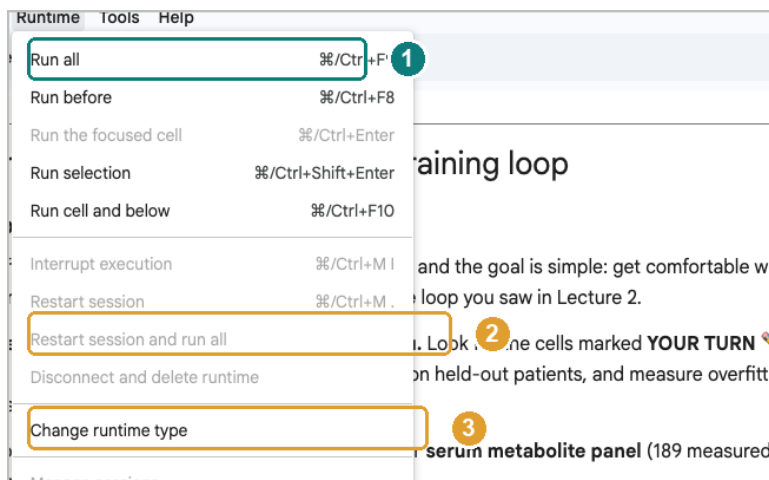
A notebook is a list of *cells*. A cell holds either text or code. Click a code cell, then press **Shift + Enter** — or hover over the (1) bracket at its left, which turns into a ► button, and click it.



- Run the cells **in order, top to bottom**. Later cells depend on earlier ones.
- While a cell runs the bracket shows a spinner; when it finishes you get a number, e.g. [1].
- **Run the first code cell, “Setup”, before anything else**. It installs the course packages and downloads the lab's data for you — about a minute, ending with `lab01 is ready`. You never download or copy data files yourself.
- Only the cells marked **YOUR TURN** (the pencil icon) need editing. Run all the others exactly as they are.

## Step 4 — If something gets tangled

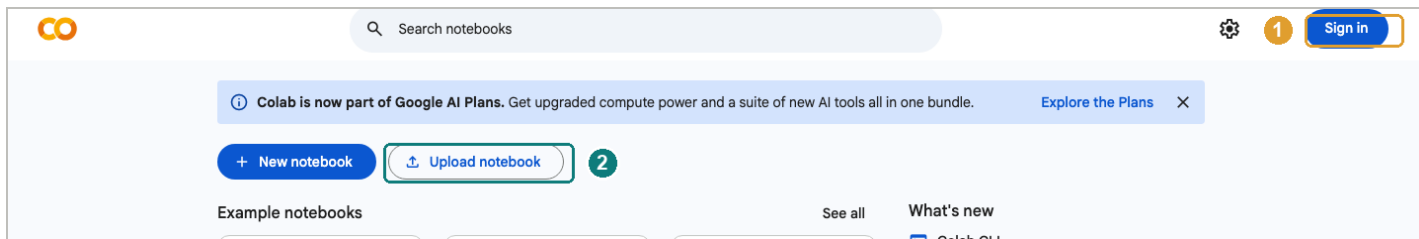
The **Runtime** menu is the fix for almost everything.



- (1) **Run all** — runs the whole notebook from the top. Good after you fix a cell.
- (2) **Restart session and run all** — throws away every variable and starts clean. This is the answer to “it worked a minute ago”. You lose nothing but the running state; your typed code stays.
- (3) **Change runtime type** — where you would pick a **T4 GPU**. **Lab 1 does not need one**; the default CPU is plenty and connects faster. You will use this in Lab 2.

## Step 5 — If the link does not open

Go to [colab.research.google.com](https://colab.research.google.com) directly, sign in (1), then use **File** → **Open notebook** → **GitHub** and paste `indigobio/msacl_dl_course`. Or use **Upload notebook** (2) if an instructor hands you the `.ipynb` file on a USB stick.



## When things go wrong

| What you see                                        | What to do                                                                                                                                                                                   |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| “Sign in” keeps reappearing, or Colab will not load | Your institutional Google account may block Colab. Sign in with a personal <code>@gmail.com</code> instead.                                                                                  |
| <code>NameError / ... is not defined</code>         | You skipped a cell, or ran them out of order. <b>Runtime</b> → <b>Restart session and run all</b> .                                                                                          |
| A cell has been running for minutes                 | Normal for training cells. If it is stuck, <b>Runtime</b> → <b>Interrupt execution</b> , then restart and run all.                                                                           |
| “Cannot connect to runtime” / disconnected          | Free Colab disconnects after a while idle. Click <b>Connect</b> and <b>Runtime</b> → <b>Run all</b> ; you lose only the running state.                                                       |
| An <code>assert</code> cell failed                  | That is the notebook checking your answer — it is meant to catch you. Read the message, fix the marked line, re-run that cell. The <b>Code Hint Sheet</b> lists the options for every blank. |
| You edited the wrong tab and lost your work         | You were in the read-only original. Redo <b>File</b> → <b>Save a copy in Drive</b> and check the tab title starts with <b>Copy of</b> .                                                      |

*Screenshots are of the Google Colab interface, captured 22 September 2026, with callout numbers added for this sheet. Colab is a Google product and its appearance changes from time to time — if a button has moved, the menu path in the text is still the one to follow. A clickable version of this sheet is [lab01\\_colab\\_setup.html](#).*